❑ READING AND MEANING OF FILE STESTLOG.CSV

A	0	1	2	3	4	5	6	7	8	9	10	Registration Number
B	14/03/2017	14/03/2017	14/03/2017	14/03/2017	14/03/2017	14/03/2017	14/03/2017	14/03/2017	14/03/2017	14/03/2017	14/03/2017	Data
C	00:00:26	01:00:26	02:00:26	03:00:26	04:00:26	05:00:26	06:00:26	07:00:26	08:00:26	09:00:26	10:00:26	Hour
D	0	0	0	0	0	0	0	0	0	0	0	Error code in hexadecimal format (0 = no error)
E												The temperature unit (degrees F or C) (#) (*)
F	28	28	28	28	28	28	28	28	29	30	30	CPU temperature
G												The unit of voltage CPU (V) (#)
H	+0.006	+0.006	+0.006	+0.006	+0.006	+0.006	+0.006	+0.006	+0.006	+0.006	+0.006	Voltage measured on the electrode E1
I												The unit of voltage E1 (V) (#)
J	+0.008	+0.008	+0.008	+0.008	+0.008	+0.008	+0.008	+0.008	+0.008	+0.008	+0.008	Reference voltage electrode E2
K												The unit of voltage E2 (V) (#)
L	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002	Differential voltage E1-E2
M												The unit of voltage (V) (#)
N	+0.007	+0.007	+0.007	+0.007	+0.007	+0.007	+0.007	+0.007	+0.007	+0.007	+0.007	Common mode voltage (E1 + E2) / 2
O												The unit of resistance (ohm) (#)
P	6.9	6.7	6.7	6.7	6.6	6.7	6.7	6.6	7.0	6.7	6.7	Resistance measured between E1 and the common
Q												The unit of resistance (ohm) (#)
R	6.9	6.7	6.8	6.7	6.5	6.8	6.8	6.6	7.0	6.8	6.7	Resistance measured between E2 and the common
S												The unit of voltage (V) (#)
T	0.005	0.006	0.006	0.005	0.005	0.005	0.005	0.004	0.004	0.004	0.004	Common mode noise at low frequency
U												The unit of voltage (V) (#)
V	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	Differential mode noise at low frequency
W												The unit of voltage (mV) (#)
X	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	Mode ADC noise at low frequency differential
Y												The unit of voltage (mV) (#)
Z	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	Mode ADC noise high frequency differential
AA												The unit of voltage (V) (#)
AB	+10.2	+10.2	+10.2	+10.2	+10.2	+10.2	+10.2	+10.2	+10.2	+10.2	+10.2	Analog circuitry positive supply voltage
AC												The unit of voltage (V) (#)
AD	-10.1	-10.2	-10.1	-10.1	-10.1	-10.2	-10.1	-10.2	-10.1	-10.2	-10.2	Negative supply voltage analog circuits
AE												The unit of current (mA) (#)
AF	24.08	24.08	24.08	24.09	24.09	24.08	24.08	24.08	24.08	24.08	24.08	Excitation current of the coils
AG												The unit of resistance (ohm) (#)
AH	137.5	137.6	137.4	137.4	137.4	137.3	137.3	137.3	137.4	137.9	138.5	Measurement of the sensor coil resistance
AI												The temperature unit (degrees F or C) (#) (*)
AJ	13.7	13.7	13.4	13.4	13.4	13.2	13.2	13.2	13.5	14.3	15.3	Temperature of the sensor coils
AK												The unit of current (mA) (#)
AL	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	The coil leakage current (insulation fault)
AM												Unit of measure of time (ms) (#)
AN	6.40	6.41	6.38	6.41	6.40	6.41	6.40	6.40	6.43	6.44	6.40	Rise time current phase A
AO												Unit of measure of time (ms) (#)
AP	6.48	6.47	6.48	6.46	6.48	6.46	6.47	6.47	6.48	6.50	6.48	Rise time current phase B

NOTE:
(#): The units are registered only if the appropriate function of the DATA LOGGER is active. Otherwise the field is empty.
(*): The temperature values can be expressed in degrees F or C, depending on the drive configuration

Standard and internal check to the instrument limits

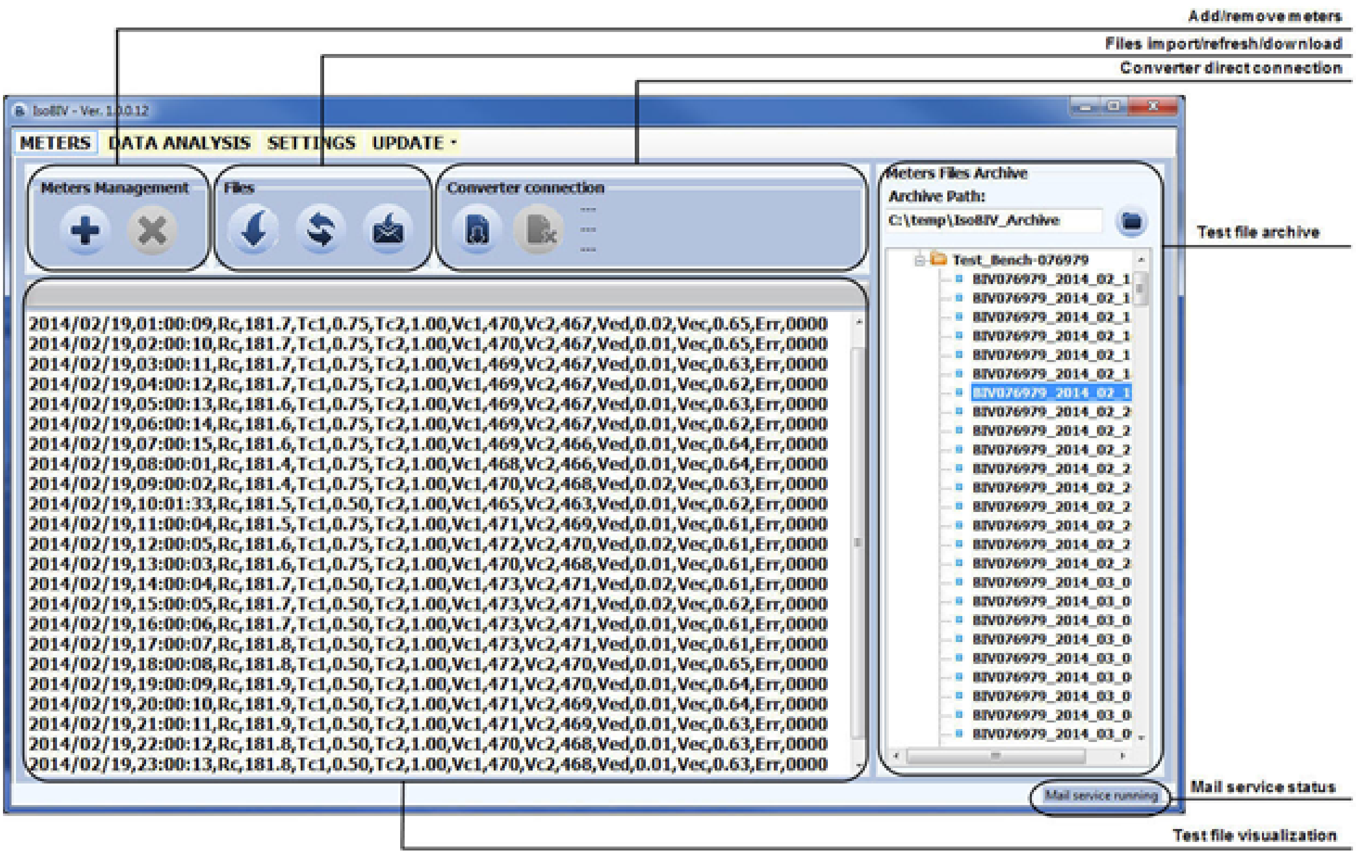
The measured data are compared with the reference values previously stored. The variation of different variable measured,shall be within the following range:

- Coil temperature (using resistance reading): within limits compatible with the lining material
- Current up times: change% detected resistance coils + 10% (tolerance range)
- Resistance between electrodes and common: between 0.3 and 3.0 times the reference early strength
- Leakage current (insulation test): less than 0.1 mA

If the values deviate beyond these limits it is generated and displayed a coded alarm.
The alarm remains active and visible on the display until next test (max. 1 hour).

SOFTWARE ISOBIV

The IsoBiv software allows analysis and processing of STESTLOG.CSV file data.



For further information, refer to the manual of ISOBIV software.

ALARM MESSAGES (CAUSES AND ACTIONS TO BE TAKEN)

MESSAGE	CAUSES	ACTION TO TAKE
NO ALARMS	All works regularly	---
[000] SYSTEM RESTART	---	---
[001] INTERNAL PS FAIL	Internal supply voltage error	Contact the service
[002] CLOCK NOT SET	System Clock not set	Set the system clock from the converter menu 13 (see also MCP function).
[003] SD CARD FAILURE	SD card not found or unreadable	check and/or replace SD card
[005] F-RAM ERROR	Error writing / reading Flash-RAM	Contact the service
[006] EXCITATION ERROR	The excitation of the sensor coils resulting from cable is interrupted	Check the connecting cables to the sensor.
[007] SIGNAL ERROR	The measure is strongly effected by external noise or the cable connecting the converter to the sensor is broken.	Check the status of the cables connecting the sensor, the grounding connections of the devices and the possible presence of noise sources.
[008] PIPE EMPTY	The measuring pipe is empty or the detection system has not been properly calibrated.	Check whether the pipe is empty or repeat the empty pipe calibration procedure.
[009] FLOW>MAX+	The flow rate is higher than the maximum positive threshold set.	Check the maximum positive flow rate threshold set and the process conditions.
[010] FLOW>MAX-	The flow rate is higher than the maximum negative threshold set.	Check the maximum negative flow rate threshold set and the process conditions.
[011] FLOW<MIN+	The flow rate is lower than the minimum positive threshold set.	Check the minimum positive flow rate threshold set and the process conditions.
[012] FLOW<MIN-	The flow rate is lower than the minimum negative threshold set.	Check the minimum negative flow rate threshold set and the process conditions.
[013] FLOW>FULL SCALE+	The flow rate is higher than the full scale positive value set on the instrument.	Check the full scale positive value set on the instrument and the process conditions.
[014] FLOW>FULL SCALE-	The flow rate is higher than the full scale negative value set on the instrument.	Check the full scale negative value set on the instrument and the process conditions.
[015] PULSE1>RANGE	The pulse generation output 1 of the device is saturated and cannot generate the sufficient number of impulses.	Set a bigger unit of volume or, if the connected counting device allows it, reduce the pulse duration value.
[016] PULSE2>RANGE	The pulse generation output 2 of the device is saturated and cannot generate the sufficient number of impulses.	Set a bigger unit of volume or, if the connected counting device allows it, reduce the pulse duration value.
[017] CALIBR.ERROR	Calibration Error	Contact the service
[018] SYSTEM FREQ.ERR	System Freq. Error	Contact the service
[019] B.DATA NOT INIT	Uninitialized data system	Contact the service
[020] FL.SENSOR ERROR	Flow rate sensor error	Contact the service
[021] BATTERY LOW	(Rechargeable) battery depleted	Contact the service to Replace the battery
[022] BATTERY V>MAX	Battery voltage (rechargeable)> max. Allowed	Contact the service to Replace the battery
[023] BATTERY I>MAX	Battery charge current> max. allowed	Contact the service to Replace the battery
[024] MAIN PS V.ERR	Main supply voltage (+ 5V) out of tolerance.	Contact the service
[025] USB VOLTAGE ERR	Voltage of USB connection out of tolerance.	Contact the service
[026] SDC ALMOST FULL	SD card space <500 MB.	For more information see function "12.9" page30.
[027] SDC FULL	SD card out of memory	Memory Full. You can not save logger. Contact the service to replace the SD memory.
[028] BATT.TEMP.CRIT	The battery can not be charged. The temperature is out of range (detected temperature <0 C° or temperature >50°)	Wait for the normal temperature reset. View Environmental Use Conditions "Environmental Use Conditions" page5.
[030] BATCHING ERROR	This error activates in two conditions (see "BATCHING" page 45)	Check the status of the actuator and the system

ERROR CODE TEST SYSTEM OF SENSOR

The codes are in hexadecimal format, the meaning is given for each bit. There are several possible error simultaneous combinations (more bits active) then that will give the combined numerical codes.

CODE	ANOMALIES DESCRIPTION	ACTION TO TAKE
0000	NO ERROR	---
0001	SENSOR TEST INSULATION: Generator power too low	Contact the service
0002	SENSOR TEST INSULATION: Generator power too high	
0004	SENSOR TEST INSULATION: Phase 1 generator voltage too low	
0008	SENSOR TEST INSULATION: Phase 1 generator voltage too high	
0010	SENSOR TEST INSULATION: Phase 1 terminal voltage coils 1 too low	
0020	SENSOR TEST INSULATION: Phase 1 terminal voltage coils 2 too low	
0040	SENSOR TEST INSULATION: Phase 2 generator voltage too low	
0080	SENSOR TEST INSULATION: Phase 2 generator voltage too high	
0100	SENSOR TEST INSULATION: Phase 2 terminal voltage coils 1 too low	
0200	SENSOR TEST INSULATION: Phase 2 terminal voltage coils 2 too low	
0400	SENSOR TEST INSULATION: Insulation loss, leakage current out of tolerance	Check: ❑ wiring between sensor converter ❑ conditions of use ❑ set parameters If the problem persists contact the service
0800	TEST TEMPERATURE (RESISTANCE) COILS: Temperature (resistance) out of tolerance	
1000	TEST TIME GETTING ON CURRENT PHASE (A): Value out of tolerance	
2000	TEST TIME GETTING ON CURRENT PHASE (B): Value out of tolerance	
4000	TEST RESISTANCE INPUTS ELECTRODES: Input value 1 out of tolerance	
8000	TEST RESISTANCE INPUTS ELECTRODES: Input value 1 out of tolerance	

At the end of its lifetime, this product shall be disposed of in full compliance with the environmental regulations of the state in which it is located.

MANUAL REVIEWS

REVIEW	DATE	DESCRIPTION
210_EN_IT_R0_1.00.0	31-01-2020	First edition
210_EN_IT_R0_1.02.0	03-07-2020	Software update, batching function added

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